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Summary

Endocrine glands: secrete hormones which diffuse into blood and induce the target organs. They are chemical messengers or organic catalysts which interact with receptor in the target organs.

Hormones speed up or slow down or alter the activities of target organs. The hypo or hyper secretion of hormones leads to serious effects on human beings. Hormones coordinate different physical and mental activities to maintain homeostasis.

Hypothalamus interlinks nervous system and endocrine system. It is located in the diencephalon of cerebrum and controls the pituitary secretion. Pituitary gland secretes six tropic hormones which regulates various physiological functions of our body. Posterior pituitary gland secretes vasopressin that regulates water and electrolyte balance. Oxytocin helps during child birth. Melatonin secreted by pineal gland regulates circadian rhythm of our body. The thyroid gland secretes thyroxine which stimulates the nervous system, skeletal growth, and regulates basal metabolic rate.

Parathyroid gland regulates calcium level in our body. Thymus gland plays a vital role in cell mediated immunity by promoting T lymphocytes maturation. Pancreas regulates blood glucose homeostasis through its secretion of insulin and glucagon.

Adrenal cortex secretes mineralocorticoids which regulates mineral metabolism, glucocorticoids regulates carbohydrate metabolism. Adrenal medulla secretes the hormones adrenalin and noradrenalin. In male, reproductive functions are controlled by testosterone secreted by the testis. In female, ovary secretes three hormones oestrogen, progesterone and relaxin that regulates reproductive functions.

Hormonal deficiency causes serious harmful effects in human. It alters physiological and biochemical functions of the body. This leads into various disorders like acromegaly, dwarfism, tetany, diabetes etc.

Opportunities to Endocrinologists

An endocrinology career is a medical career that involves studying hormones and their effects on human body. They investigate and find new ways of treatment of hormonal imbalance. Just like doctors, endocrinologists speak with patients about their medical history and share current findings. They also study the test result of patients and advise them for treatment.

After completing basic degree in medicine (MBBS) doctors can study super specialty course in endocrinology such as endocrine surgery, general endocrinology, paediatric endocrinology etc.

Glossary

Alternation of generation – Alternation of haploid sexual and diploid asexual generation in the life cycle of an animal.

Autonomy - Breaking of a body part.

Dioecious - Animals in which male and female reproductive organs occur in separate individuals.

Acidosis – condition characterised by lower blood pH, due to the increase of keto acids (ketosis)

cAMP – Cyclic adenosine mono phosphate acts as a second messenger (intracellular messenger) in the case of peptide hormone.